

```

In[1]:= If[! TrueQ[Private`$ev], Block[{Private`$ev = True},
  Print["Total time: ", First@AbsoluteTiming[ClearAll["Global`*"];
    NotebookEvaluate[EvaluationNotebook[], InsertResults -> True]], " s"]]]

(* =====*)
(*Verification of low Mach number expansion for u and φ up to 8th order*)
(*All equations should yield {0,0} or 0. If all pass,prints True.*)
(* =====*)

(*_ 0th order _*)
u0 = {(3 Tf) / (4 (3 μ + α)) (2 μ + α - α x2^2), 0};

(*_ 2nd order _*)
φ2 = (3 Tf (2 - x1) α μ) / (2 (α + 3 μ));

u2 = {1 / (320 (α + 3 μ)^3) × 9 Tf^2 α
  (Tf x2^4 α ((-5 + x2^2) α - 10 μ) + 40 (-2 + x1) μ (α - x2^2 α + 2 μ) (α + 3 μ)), 0};

(*_ 4th order _*)
φ4 = - (1 / (32 (α + 3 μ)^3)) 9 Tf^2 α μ (3 Tf x1 (α + 2 μ)^2 +
  4 (α + 3 μ) ((-1 + (-4 + x1) x1 + x2^2) α μ + (-1 + x2^2) α ν - 2 μ (μ + ν)));

u4 = {- (1 / (358400 (α + 3 μ)^5)) 27
  Tf^3 α (Tf^2 x2^4 α ((1020 - 70 x2^2 - 75 x2^4 + 8 x2^6) α^3 -
    10 (-623 + 28 x2^2 + 15 x2^4) α^2 μ - 280 (-45 + x2^2) α μ^2 + 8400 μ^3) -
    80 Tf μ (α + 3 μ) ((111 x1 - 105 x1 x2^2 - 70 (-2 + x1) x2^4 + 14 (-2 + x1) x2^6) α^3 -
    28 (-23 x1 + 15 x1 x2^2 + 5 (-2 + x1) x2^4) α^2 μ -
    420 x1 (-3 + x2^2) α μ^2 + 840 x1 μ^3) +
    5600 μ (α + 3 μ)^2 (μ ((-14 - 6 (-4 + x1) x1 + 6 (2 + (-4 + x1) x1) x2^2 - x2^4) α^2 -
    4 (3 (-4 + x1) x1 + 2 (3 + x2^2)) α μ + 8 μ^2) + 2 (α + 2 μ) (α - 2 x2^2 α + 2 μ) ν)),
  - (27 Tf^4 x2 α^3 μ ((6 - 7 x2^4 + x2^6) α - 14 (-1 + x2^4) μ)) /
  (4480 (α + 3 μ)^4)};

(*_ 6th order _*)
φ6 = 1 / (17920 (α + 3 μ)^5) × 27 Tf^3 α μ
  (-3 Tf^2 x1 (α + 2 μ) (111 α^3 + 644 α^2 μ + 1260 α μ^2 + 840 μ^3) -
  28 Tf (α + 3 μ) (15 ((-1 + x1 (-8 + 3 x1) + x2^2) α - 2 μ) μ (α + 2 μ)^2 +
  (- (15 - 15 x2^2 - 5 x2^4 + x2^6) α^3) +
  10 (-9 + 6 x2^2 + x2^4) α^2 μ + 60 (-3 + x2^2) α μ^2 - 120 μ^3) ν) -
  1120 α μ (α + 3 μ)^2 ((6 - 6 x1^2 + x1^3 + 3 (-2 + x1) x2^2) α μ +
  (6 - 5 x1 + 3 (-2 + x1) x2^2) α ν - 2 μ (-6 μ + 8 x1 μ - 6 ν + 5 x1 ν)));

u6 = {1 / (11480268800 (α + 3 μ)^7) × 27 Tf^4 α
  (19219200 Tf x1^2 α^3 μ^2 (α + 3 μ)^2 ((6 - 35 x2^4 + 7 x2^6) α + 14 (1 - 5 x2^4) μ) -
  16016 x1 α^2 μ (α + 3 μ) (4800 Tf α μ (α + 3 μ) ((6 - 35 x2^4 + 7 x2^6) α + 14 (1 - 5 x2^4)
  μ) + Tf^2 ((-2879 + 15030 x2^4 + 420 x2^6 - 1845 x2^8 + 198 x2^10) α^3 -
  2 (9197 - 46410 x2^4 - 840 x2^6 + 1845 x2^8) α^2 μ +
  120 (-317 + 1575 x2^4 + 14 x2^6) α μ^2 + 25200 (-1 + 5 x2^4) μ^3) -
  13440 (-1 + 5 x2^4) α μ (α + 3 μ)^2 (3 μ + 2 ν)) + 48048 (2 - x1) α
  μ (α + 3 μ) (Tf^2 x2^4 α ((1020 - 70 x2^2 - 75 x2^4 + 8 x2^6) α^3 -

```

$$\begin{aligned}
& 10 (-623 + 28 x^2 + 15 x^2^4) \alpha^2 \mu - 280 (-45 + x^2) \alpha \mu^2 + 8400 \mu^3 - \\
& 80 \text{Tf} \mu (\alpha + 3 \mu) ((111 x - 105 x^2 - 70 (-2 + x) x^2 + 14 (-2 + x) x^2^6) \alpha^3 - \\
& 28 (-23 x + 15 x^2 + 5 (-2 + x) x^2^4) \alpha^2 \mu - \\
& 420 x (-3 + x^2) \alpha \mu^2 + 840 x \mu^3) + 5600 \mu (\alpha + 3 \mu)^2 \\
& (\mu ((-14 - 6 (-4 + x) x + 6 (2 + (-4 + x) x) x^2 - x^2^4) \alpha^2 - 4 (3 (-4 + x) x + \\
& 2 (3 + x^2)) \alpha \mu + 8 \mu^2) + 2 (\alpha + 2 \mu) (\alpha - 2 x^2 \alpha + 2 \mu) \nu)) + \\
& 3363360 \alpha \mu (\alpha + 3 \mu) (\text{Tf} x^2 \alpha (-5 + x^2) \alpha - 10 \mu) + 40 (-2 + x) \mu \\
& (\alpha - x^2 \alpha + 2 \mu) (\alpha + 3 \mu)) (3 \text{Tf} x (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) ((-1 + (-4 + x) x + x^2) \alpha \mu + (-1 + x^2) \alpha \nu - 2 \mu (\mu + \nu))) + \\
& 480480 \mu (\alpha - x^2 \alpha + 2 \mu) (\alpha + 3 \mu) \\
& (3 \text{Tf}^2 x (\alpha + 2 \mu) (111 \alpha^3 + 644 \alpha^2 \mu + 1260 \alpha \mu^2 + 840 \mu^3) + \\
& 28 \text{Tf} (\alpha + 3 \mu) (15 ((-1 + x) (-8 + 3 x) + x^2) \alpha - 2 \mu) \mu (\alpha + 2 \mu)^2 + \\
& (-((15 - 15 x^2 - 5 x^2^4 + x^2^6) \alpha^3) + 10 (-9 + 6 x^2 + x^2^4) \\
& \alpha^2 \mu + 60 (-3 + x^2) \alpha \mu^2 - 120 \mu^3) \nu) + \\
& 1120 \alpha \mu (\alpha + 3 \mu)^2 ((6 - 6 x^2 + x^2^3 + 3 (-2 + x) x^2) \alpha \mu + \\
& (6 - 5 x + 3 (-2 + x) x^2) \alpha \nu - 2 \mu ((-6 + 8 x) \mu + (-6 + 5 x) \nu))) + \\
& x^2 \alpha (96096 \text{Tf}^2 \alpha \mu (\alpha + 3 \mu) ((2850 + 560 x^2 - 615 x^2^4 + 66 x^2^6) \alpha^3 + \\
& 10 (1820 + 224 x^2 - 123 x^2^4) \alpha^2 \mu + 280 (135 + 8 x^2) \alpha \mu^2 + 25200 \mu^3) + \\
& \text{Tf}^3 ((-17118101 - 6774768 x^2 + 4830111 x^2^4 + 290290 \\
& x^2^6 - 236509 x^2^8 + 17028 x^2^{10}) \alpha^5 - \\
& 182 (975953 + 313764 x^2 - 162657 x^2^4 - 6380 x^2^6 + 2599 x^2^8) \alpha^4 \mu + \\
& 4004 (-183090 - 44436 x^2 + 14985 x^2^4 + 290 x^2^6) \alpha^3 \mu^2 + \\
& 1081080 (-1386 - 224 x^2 + 37 x^2^4) \alpha^2 \mu^3 - \\
& 60540480 (25 + 2 x^2) \alpha \mu^4 - 605404800 \mu^5) - \\
& 2152550400 \alpha^2 \mu^2 (\alpha + 3 \mu)^3 (3 \mu + 2 \nu) - 160160 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 \\
& (-2520 \mu^2 (12 \mu + 7 \nu) - 56 \alpha \mu ((60 + 11 x^2) \mu + 3 (105 + x^2) \nu) + \\
& \alpha^2 ((6132 - 2996 x^2 + 234 x^2^4) \mu + 3 (-1414 - 28 x^2 + 47 x^2^4) \nu))) , \\
& 1 / (716800 (\alpha + 3 \mu)^6) \times 27 \text{Tf}^4 \alpha^3 \mu (240 \text{Tf} (2 - x) x^2 \alpha \mu (\alpha + 3 \mu) \\
& ((6 - 7 x^2 + x^2^6) \alpha - 14 (-1 + x^2) \mu) - \\
& 2400 \text{Tf} x \alpha \mu (\alpha + 3 \mu) (x^2^7 \alpha - 7 x^2^5 (\alpha + 2 \mu) + 2 x^2 (3 \alpha + 7 \mu)) + \\
& x^2 (4800 \text{Tf} (-1 + x^2) \alpha \mu (\alpha + 3 \mu) ((-6 - 6 x^2 + x^2^4) \alpha - 14 (1 + x^2) \mu) + \\
& \text{Tf}^2 ((-2879 + 3006 x^2 + 60 x^2^6 - 205 x^2^8 + 18 x^2^{10}) \alpha^3 + \\
& 2 (-9197 + 9282 x^2 + 120 x^2^6 - 205 x^2^8) \alpha^2 \mu + \\
& 120 (-317 + 315 x^2 + 2 x^2^6) \alpha \mu^2 + 25200 (-1 + x^2) \mu^3) - \\
& 13440 (-1 + x^2) \alpha \mu (\alpha + 3 \mu)^2 (3 \mu + 2 \nu))));
\end{aligned}$$

(*_ 8th order _*)

$$\begin{aligned}
\varphi 8 = & - (1 / (2867200 (\alpha + 3 \mu)^7)) 81 \text{Tf}^4 \alpha \mu \\
& (\text{Tf}^3 x (12869 \alpha^6 + 142052 \alpha^5 \mu + 655868 \alpha^4 \mu^2 + 1632720 \alpha^3 \mu^3 + 2328480 \alpha^2 \\
& \mu^4 + 1814400 \alpha \mu^5 + 604800 \mu^6) + 4 \text{Tf}^2 (\alpha + 3 \mu) (-100800 \mu^5 (\mu + \nu) + \\
& 50400 \alpha \mu^4 ((-5 - 16 x + 8 x^2 + x^2^2) \mu + (-5 + x^2) \nu) + 1680 \alpha^2 \mu^3 \\
& ((-151 - 960 x + 480 x^2 + 60 x^2^2 - 15 x^2^4) \mu + (-151 + 60 x^2 + 10 x^2^4) \nu) + \\
& 40 \alpha^3 \mu^2 ((-3210 - 31164 x + 15414 x^2 + 1680 x^2^2 - 945 x^2^4 + 35 x^2^6) \mu + \\
& (-3210 + 1911 x^2 + 630 x^2^4 - 28 x^2^6) \nu) + 10 \alpha^4 \mu \\
& (2 (-1632 - 21876 x + 10626 x^2 + 870 x^2^2 - 805 x^2^4 + 70 x^2^6 - 12 x^2^8) \mu + \\
& (-3264 + 2598 x^2 + 1253 x^2^4 - 112 x^2^6 - 15 x^2^8) \nu) + \alpha^5 \\
& (2 (-1665 - 29160 x + 13860 x^2 + 675 x^2^2 - 975 x^2^4 + 175 x^2^6 - 60 x^2^8 + 4 x^2^{10} \\
& 10) \mu + (-3330 + 3330 x^2 + 2070 x^2^4 - 280 x^2^6 - 75 x^2^8 + 8 x^2^{10}) \nu)) + \\
& 320 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 (-1680 \mu^3 (-4 \mu + 13 x \mu - 4 \nu + 8 x \nu) + \\
& 140 \alpha \mu^2 ((72 - 90 x - 114 x^2 + 22 x^3 - 24 x^2^2 + 21 x^2^2) \mu +
\end{aligned}$$

$$\begin{aligned}
& 3 (24 - 48 x_1 - 8 x_2^2 + 7 x_1 x_2^2) v) + 14 \alpha^2 \mu \\
& ((-1140 x_1^2 + 220 x_1^3 + x_1 (239 + 210 x_2^2 - 35 x_2^4) + 10 (36 - 24 x_2^2 + 7 x_2^4)) \\
& \mu + (-40 (-9 + 6 x_2^2 + x_2^4) + x_1 (-733 + 210 x_2^2 + 20 x_2^4)) v) + \\
& \alpha^3 ((-3990 x_1^2 + 770 x_1^3 + 14 (60 - 60 x_2^2 + 35 x_2^4 - 7 x_2^6) + x_1 \\
& (2106 + 735 x_2^2 - 245 x_2^4 + 49 x_2^6)) \mu + (x_1 (-1770 + 735 x_2^2 + 140 x_2^4 - \\
& 28 x_2^6) + 56 (15 - 15 x_2^2 - 5 x_2^4 + x_2^6)) v) + 22400 \alpha \mu (\alpha + 3 \mu)^3 \\
& (8 \mu^2 (10 \mu^2 + 17 \mu v + 7 v^2) - 4 \alpha \mu ((28 - 160 x_1 + 45 x_1^2 + 15 x_2^2) \mu^2 + \\
& (14 - 100 x_1 + 27 x_1^2 + 27 x_2^2) \mu v + 2 (-7 + 6 x_2^2) v^2) + \\
& \alpha^2 ((-76 - 40 x_1^3 + 5 x_1^4 + 66 x_2^2 + 19 x_2^4 + 6 x_1^2 (1 + 5 x_2^2) - \\
& 8 x_1 (-32 + 15 x_2^2)) \mu^2 + (-62 + 42 x_2^2 + 23 x_2^4 - 40 x_1 (-5 + 3 x_2^2) + \\
& 6 x_1^2 (-9 + 5 x_2^2)) \mu v + 2 (7 - 12 x_2^2 + 4 x_2^4) v^2)) ;
\end{aligned}$$

$$u8 = \{1 / (22960537600 (\alpha + 3 \mu)^8) \times 81 Tf^5 \alpha$$

$$\begin{aligned}
& \mu (-6006 \alpha (Tf^2 x_2^4 \alpha ((1020 - 70 x_2^2 - 75 x_2^4 + 8 x_2^6) \alpha^3 - \\
& 10 (-623 + 28 x_2^2 + 15 x_2^4) \alpha^2 \mu - 280 (-45 + x_2^2) \alpha \mu^2 + 8400 \mu^3) - \\
& 80 Tf \mu (\alpha + 3 \mu) ((111 x_1 - 105 x_1 x_2^2 - 70 (-2 + x_1) x_2^4 + 14 (-2 + x_1) x_2^6) \alpha^3 - \\
& 28 (-23 x_1 + 15 x_1 x_2^2 + 5 (-2 + x_1) x_2^4) \alpha^2 \mu - \\
& 420 x_1 (-3 + x_2^2) \alpha \mu^2 + 840 x_1 \mu^3) + \\
& 5600 \mu (\alpha + 3 \mu)^2 (\mu ((-14 - 6 (-4 + x_1) x_1 + 6 (2 + (-4 + x_1) x_1) x_2^2 - x_2^4) \\
& \alpha^2 - 4 (3 (-4 + x_1) x_1 + 2 (3 + x_2^2)) \alpha \mu + 8 \mu^2) + \\
& 2 (\alpha + 2 \mu) (\alpha - 2 x_2^2 \alpha + 2 \mu) v)) * (3 Tf x_1 (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) ((-1 + (-4 + x_1) x_1 + x_2^2) \alpha \mu + (-1 + x_2^2) \alpha v - 2 \mu (\mu + v))) - \\
& 12012 \alpha (Tf x_2^4 \alpha ((-5 + x_2^2) \alpha - 10 \mu) + 40 (-2 + x_1) \mu (\alpha - x_2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) * \\
& (-3 Tf^2 x_1 (\alpha + 2 \mu) (111 \alpha^3 + 644 \alpha^2 \mu + 1260 \alpha \mu^2 + 840 \mu^3) - \\
& 28 Tf (\alpha + 3 \mu) (15 ((-1 + x_1 (-8 + 3 x_1) + x_2^2) \alpha - 2 \mu) \mu (\alpha + 2 \mu)^2 + \\
& (-((15 - 15 x_2^2 - 5 x_2^4 + x_2^6) \alpha^3) + 10 (-9 + 6 x_2^2 + x_2^4) \\
& \alpha^2 \mu + 60 (-3 + x_2^2) \alpha \mu^2 - 120 \mu^3) v) - \\
& 1120 \alpha \mu (\alpha + 3 \mu)^2 ((6 - 6 x_1^2 + x_1^3 + 3 (-2 + x_1) x_2^2) \alpha \mu + \\
& (6 - 5 x_1 + 3 (-2 + x_1) x_2^2) \alpha v - 2 \mu ((-6 + 8 x_1) \mu + (-6 + 5 x_1) v))) + \\
& 6006 (\alpha - x_2^2 \alpha + 2 \mu) (Tf^3 x_1 (12869 \alpha^6 + 142052 \alpha^5 \mu + 655868 \alpha^4 \mu^2 + \\
& 1632720 \alpha^3 \mu^3 + 2328480 \alpha^2 \mu^4 + 1814400 \alpha \mu^5 + 604800 \mu^6) + \\
& 4 Tf^2 (\alpha + 3 \mu) (-100800 \mu^5 (\mu + v) + 50400 \alpha \mu^4 \\
& ((-5 - 16 x_1 + 8 x_1^2 + x_2^2) \mu + (-5 + x_2^2) v) + 1680 \alpha^2 \mu^3 ((-151 - \\
& 960 x_1 + 480 x_1^2 + 60 x_2^2 - 15 x_2^4) \mu + (-151 + 60 x_2^2 + 10 x_2^4) v) + \\
& 40 \alpha^3 \mu^2 ((-3210 - 31164 x_1 + 15414 x_1^2 + 1680 x_2^2 - 945 x_2^4 + 35 x_2^6) \\
& \mu + (-3210 + 1911 x_2^2 + 630 x_2^4 - 28 x_2^6) v) + 10 \alpha^4 \mu \\
& (2 (-1632 - 21876 x_1 + 10626 x_1^2 + 870 x_2^2 - 805 x_2^4 + 70 x_2^6 - 12 x_2^8) \\
& \mu + (-3264 + 2598 x_2^2 + 1253 x_2^4 - 112 x_2^6 - 15 x_2^8) v) + \\
& \alpha^5 (2 (-1665 - 29160 x_1 + 13860 x_1^2 + 675 x_2^2 - 975 x_2^4 + \\
& 175 x_2^6 - 60 x_2^8 + 4 x_2^{10}) \mu + \\
& (-3330 + 3330 x_2^2 + 2070 x_2^4 - 280 x_2^6 - 75 x_2^8 + 8 x_2^{10}) v)) + \\
& 320 Tf \alpha \mu (\alpha + 3 \mu)^2 (-1680 \mu^3 (-4 \mu + 13 x_1 \mu - 4 v + 8 x_1 v) + \\
& 140 \alpha \mu^2 ((72 - 90 x_1 - 114 x_1^2 + 22 x_1^3 - 24 x_2^2 + 21 x_1 x_2^2) \mu + 3 \\
& (24 - 48 x_1 - 8 x_2^2 + 7 x_1 x_2^2) v) + 14 \alpha^2 \mu ((-1140 x_1^2 + 220 x_1^3 + \\
& x_1 (239 + 210 x_2^2 - 35 x_2^4) + 10 (36 - 24 x_2^2 + 7 x_2^4)) \mu + \\
& (-40 (-9 + 6 x_2^2 + x_2^4) + x_1 (-733 + 210 x_2^2 + 20 x_2^4)) v) + \\
& \alpha^3 ((-3990 x_1^2 + 770 x_1^3 + 14 (60 - 60 x_2^2 + 35 x_2^4 - 7 x_2^6) + \\
& x_1 (2106 + 735 x_2^2 - 245 x_2^4 + 49 x_2^6)) \mu + (x_1 (-1770 + 735 x_2^2 + \\
& 140 x_2^4 - 28 x_2^6) + 56 (15 - 15 x_2^2 - 5 x_2^4 + x_2^6)) v) + 22400 \alpha \\
& \mu (\alpha + 3 \mu)^3 (8 \mu^2 (10 \mu^2 + 17 \mu v + 7 v^2) - 4 \alpha \mu ((28 - 160 x_1 + 45 x_1^2 + 15
\end{aligned}$$

_____*)

(*Verification of equations 1-8*)

(*=====*)

(*Equation 1*)

```
eq1 = Grad[φ2, {x1, x2}] - μ Laplacian[u0, {x1, x2}] // FullSimplify;
test1 = (eq1 == {0, 0});
```

(*Equation 2*)

```
eq2 = Div[u2, {x1, x2}] + Grad[φ2, {x1, x2}].u0 // FullSimplify;
test2 = (eq2 == 0);
```

(*Equation 3*)

```
eq3 = Grad[φ4, {x1, x2}] -
      μ Laplacian[u2, {x1, x2}] - (μ + ν) Grad[Div[u2, {x1, x2}], {x1, x2}] +
      Grad[u0, {x1, x2}].u2 + Grad[u2, {x1, x2}].u0 // FullSimplify;
test3 = (eq3 == {0, 0});
```

(*Equation 4*)

```
eq4 = Div[u4, {x1, x2}] + Grad[φ2, {x1, x2}].u2 +
      Grad[φ4, {x1, x2}].u0 + φ2 Div[u2, {x1, x2}] // FullSimplify;
test4 = (eq4 == 0);
```

(*Equation 5*)

```
eq5 = Grad[φ6, {x1, x2}] - μ Laplacian[u4, {x1, x2}] - (μ + ν)
      Grad[Div[u4, {x1, x2}], {x1, x2}] + Grad[u0, {x1, x2}].u4 + Grad[u4, {x1, x2}].u0 +
      φ2 Grad[u0, {x1, x2}].u2 + φ2 Grad[u2, {x1, x2}].u0 // FullSimplify;
test5 = (eq5 == {0, 0});
```

(*Equation 6*)

```
eq6 = Div[u6, {x1, x2}] + Grad[φ2, {x1, x2}].u4 + Grad[φ4, {x1, x2}].u2 +
      Grad[φ6, {x1, x2}].u0 + φ4 Div[u2, {x1, x2}] + φ2 Div[u4, {x1, x2}] // FullSimplify;
test6 = (eq6 == 0);
```

(*Equation 7-momentum for φ8*)

```
eq7 = Grad[φ8, {x1, x2}] - μ Laplacian[u6, {x1, x2}] - (μ + ν)
      Grad[Div[u6, {x1, x2}], {x1, x2}] + (Grad[u6, {x1, x2}].u0 + Grad[u0, {x1, x2}].u6) +
      (Grad[u4, {x1, x2}].u2 + Grad[u2, {x1, x2}].u4) +
      φ2 (Grad[u4, {x1, x2}].u0 + Grad[u2, {x1, x2}].u2 + Grad[u0, {x1, x2}].u4) +
      φ4 (Grad[u2, {x1, x2}].u0 + Grad[u0, {x1, x2}].u2) // FullSimplify;
test7 = (eq7 == {0, 0});
```

(*Equation 8-continuity for u8*)

```
eq8 = Div[u8, {x1, x2}] + Grad[φ2, {x1, x2}].u6 + Grad[φ4, {x1, x2}].u4 +
      Grad[φ6, {x1, x2}].u2 + Grad[φ8, {x1, x2}].u0 + φ6 Div[u2, {x1, x2}] +
      φ4 Div[u4, {x1, x2}] + φ2 Div[u6, {x1, x2}] // FullSimplify;
test8 = (eq8 == 0);
```

(*_ Boundary checks: v=0 at x2=±1 _*)

```
b2 = {FullSimplify[u2[[2]] /. x2 → 1] == 0, FullSimplify[u2[[2]] /. x2 → -1] == 0};
b4 = {FullSimplify[u4[[2]] /. x2 → 1] == 0, FullSimplify[u4[[2]] /. x2 → -1] == 0};
```

```

b6 = {FullSimplify[u6[[2]] /. x2 → 1] == 0, FullSimplify[u6[[2]] /. x2 → -1] == 0};
b8 = {FullSimplify[u8[[2]] /. x2 → 1] == 0, FullSimplify[u8[[2]] /. x2 → -1] == 0};
allBdry = And@@Flatten[{b2, b4, b6, b8}];

```

```

(*=====*)
(*Output section*)
(*=====*)
Print["====="];
Print["Verification up to 8th order (x1,x2)"];
Print["====="];
Print["eq1: ", test1];
Print["eq2: ", test2];
Print["eq3: ", test3];
Print["eq4: ", test4];
Print["eq5: ", test5];
Print["eq6: ", test6];
Print["eq7: ", test7];
Print["eq8: ", test8];
Print["All main equations passed: ",
      test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8];

Print["-----"];
Print["Boundary conditions (v = 0 at x2 = ±1):"];
Print[" u2: ", b2];
Print[" u4: ", b4];
Print[" u6: ", b6];
Print[" u8: ", b8];
Print["All boundary tests passed: ", allBdry];

Print["-----"];
Print["Overall status: ",
      test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8 && allBdry];
Print["====="];
Total time: 0.250022 s

=====

Verification up to 8th order (x1,x2)

=====

eq1: True
eq2: True
eq3: True
eq4: True
eq5: True
eq6: True
eq7: True
eq8: True
All main equations passed: True

```

Boundary conditions ($v = 0$ at $x_2 = \pm 1$):

u2: {True, True}

u4: {True, True}

u6: {True, True}

u8: {True, True}

All boundary tests passed: True

Overall status: True
=====